

AKANKSHA SAMINDLA

DATA ANALYTICS ENGINEER

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SUMMARY

Data Analytics Engineer with 5 years of experience designing and implementing cloud-native ETL, streaming, and analytical data solutions across AWS, Azure, and Databricks environments. Proficient in building large-scale data pipelines using **AWS Glue Jobs, PySpark, and Databricks**, integrating complex sources to deliver real-time insights and data reliability. Expert in developing robust, automated, and compliant architectures leveraging **Lambda, Kinesis, Snowflake, Airflow, and dbt**—driving observability, performance optimization, and regulatory compliance (HIPAA, GDPR). Skilled in **data modeling, orchestration, and CI/CD automation** using Terraform, Docker, and GitHub Actions, ensuring consistency and scalability across environments. Certified **AWS Cloud Practitioner** and **Tableau Desktop Specialist** with a strong record of cross-functional collaboration, enabling data-driven decision-making and operational efficiency in healthcare and financial domains.

TECHNICAL SKILLS

Programming & Scripting:	Python, SQL, R, Shell scripting, Java
Data Formats & Modeling:	JSON, XML, Avro, Parquet; Snowflake/Star schemas, Medallion architecture
Data Eng. & Orchestration:	ETL pipelines, Change Data Capture (CDC), Apache Airflow, AWS Glue Jobs, PySpark, Databricks, dbt, Hive, SparkSQL, SSIS, SSRS
Cloud Platforms & Services:	AWS (S3, Glue, Lambda, Kinesis, Redshift, EC2, CloudWatch, CloudTrail) Azure (Data Factory, Synapse Analytics, Data Lake Gen2, Blob Storage, Azure Fabric) GCP (BigQuery, Dataflow)
Data Warehousing & Databases:	Snowflake, Amazon Redshift, Azure Synapse, PostgreSQL, MySQL, MongoDB, Oracle
Machine Learning & AI Frameworks:	Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Keras, DVC, MLflow
ML Techniques:	Supervised & unsupervised learning, regression, clustering, time series forecasting, feature engineering, A/B testing, model deployment
Visualization & Reporting Tools:	Tableau, Power BI, AWS QuickSight, Excel
DevOps, Infrastructure & CI/CD:	Git, Jenkins, Docker, Kubernetes, GitHub Actions, Terraform, CloudFormation
Data Governance & Compliance:	HIPAA, GDPR, CCPA, role-based access control (RBAC), data masking, encryption (KMS)

EDUCATION

University of Houston USA	Aug 2023 – May 2025
Masters in Engineering data science	
CMR College of Engineering and Technology India	Jul 2017 – May 2021
Bachelor of Technology in Electronics and Instrumentation Engineering	

PROFESSIONAL EXPERIENCE

University of Houston	USA
Data Analytics Engineer	May 2024 – Present
- Architected scalable end-to-end AWS Glue, Lambda, Kinesis pipelines integrating Apache Spark Streaming and NiFi, demonstrating problem-solving skills to process 10,000+ IoT sensors daily, improving data throughput by 30%. - Developed and orchestrated PySpark-based AWS Glue Jobs for large-scale data transformations and schema validation, optimizing job runtimes by 40% and enabling real-time analytics for IoT workloads. Automated schema enforcement and version control leveraging dbt and AWS Glue Catalog, showcasing attention to detail and collaboration, saving over 15 manual hours monthly while boosting pipeline reliability by 25%. - Engineered CDC ingestion workflows from PostgreSQL and MongoDB to Snowflake, applying data partitioning and native cloning techniques to reduce refresh cycles by hours, accelerating analytics delivery. - Implemented robust data lineage, monitoring, and auditing via AWS CloudTrail, CloudWatch, and Elasticsearch, ensuring HIPAA and GDPR compliance through effective cross-team communication. Orchestrated cloud infrastructure provisioning and deployments using Terraform and GitHub Actions, driving operational efficiency by slashing deployment time from days to under four hours. - Administered RBAC and row-level security policies in Snowflake, collaborating closely with Agile teams to safeguard sensitive healthcare data for 50+ users. Optimized AWS S3 data layouts and Athena federated queries through fine-grained partitioning strategies, reducing query runtimes by 12 minutes and enhancing anomaly detection speed by 40%.	
Capgemini	India
Data Engineer/Analyst	Jul 2021 – Jul 2023
- Designed and maintained robust Apache Airflow DAGs on AWS to orchestrate sensitive healthcare and finance batch workloads, reducing job failures by 12 weekly incidents and maintaining high data quality standards for regulated environments. - Built and deployed PySpark ETL scripts as AWS Glue Jobs to transform multi-source healthcare and finance data, achieving 35% faster data ingestion and improved schema consistency. Developed scalable real-time ingestion pipelines for healthcare and financial analytics use cases, leveraging PostgreSQL, APIs, Kafka, and ETL tools to achieve sub-5-second analytics delivery latency and process over 1 million messages daily.	

- Established automated data quality **frameworks** with **AWS Glue** and **Lambda**, meeting **HIPAA** and **finance** compliance **requirements**, cutting manual error investigations by **20 hours** monthly and boosting **stakeholder confidence**.
- **Streamlined SQL** transformation workflows with **dbt** and **AWS Glue Catalog** on **regulatory data**, shortening code review cycles from **3 days to 1 day** and decreasing bugs by **30%**.
- Tuned **PostgreSQL** and **MS SQL Server** on **AWS RDS** to optimize reporting for banking and **healthcare data systems**, improving query response time from **8 to 6 seconds** and supporting higher throughput during critical peak loads.
- **Orchestrated** deployment of **Docker containers** and **Kubernetes microservices** on **AWS EKS** for highly secure **healthcare** and **finance platforms**, collaborating with **DevOps** teams to scale system capacity for **3x more** concurrent users and reduce downtime by **40%**.
- Automated environment provisioning using **Terraform**, accelerating secure project setup from **2 days to 12 hours** and ensuring **100% configuration consistency** in regulated **industry projects**.
- Collaborated cross-functionally in Agile sprints with business analysts and compliance teams to deliver **15+ new** features for healthcare and **financial applications**, **quickly** resolving bugs and enhancing platform stability.
- Implemented centralized monitoring with **AWS CloudWatch**, **Prometheus**, and **Grafana**, providing real-time incident alerts and compliance visibility with detection times under 3 minutes.
- **Led data governance** and **RBAC** process improvements that enabled audit readiness, **achieving zero** compliance issues during **healthcare** and **finance** regulatory **reviews**.

KPMG

Intern Data Engineer

India

Jul 2020 – Jul 2021

- **Developed fraud detection pipelines** with Azure Data Factory and Databricks (PySpark, SQL), **applying feature engineering on millions of records, resulting in a 15% uplift in fraud detection accuracy**.
- Constructed Medallion architecture layers in **Azure Data Lake Gen2** with **Delta Lake**, **shaping clean**, governed data for audits and **analytics** with **99%** accuracy through collaborative efforts.
- Built streaming ingestion pipelines using **Azure Event Hubs** and **Spark Structured Streaming**, enabling real-time fraud alerts and reducing risk exposure time by **30%**, highlighting responsiveness and **problem-solving**.
- Automated **CI/CD deployments** of **Azure Synapse SQL pools** and **Power BI** dashboards with **GitHub Actions**, increasing deployment speed by **40%** and promoting efficiency in **team workflows**.
- Established **HIPAA-compliant monitoring, logging, RBAC**, and data masking using **Azure Monitor** and **Log Analytics**, protecting over **100,000** sensitive records with attention to security best practices and **teamwork**.

PROJECTS

Missing Child Identification System – Python, OpenCV, TensorFlow, ResNet50

- Built a **CNN-based** model achieving **90%+ accuracy** in detecting missing children using **5,000+** processed images.
- Enabled real-time image ingestion from **CCTV** and **external uploads** with law enforcement integration

Allstate Business Insurance Expert Chatbot (ABIE) – NLP, Dialogflow, JSON

- Transformed **10,000+ DITA XML** topics into optimized **JSON** for **real-time chatbot** use.
- Implemented **tracking dashboards** and **metadata** enrichment workflows for performance **enhancement**.

CERTIFICATION

- **AWS Cloud Practitioner**
- **Tableau Desktop Specialist**